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sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_994: Solar Calibration g_eff Convergence

Abstract

We derive the effective solar surface gravity including the 26-layer buoyancy correction:

$$g_{\text{eff}} = \frac{g_N}{1 + \frac{\beta_i \cdot S_{26}}{[\text{SSq}] \cdot 13.5}}$$

With $g_N = GM_{\odot}/R_{\odot}^2 = 274 \text{ m/s}^2$, $\beta_i = 0.603$, $S_{26} \approx 19.5$, $[\text{SSq}] = 0.57$:

$$g_{\text{eff}} = \frac{274}{1 + \frac{0.603 \times 19.5}{0.57 \times 13.5}} = \frac{274}{1 + 1.529} = \frac{274}{2.529} \approx 108.05 \text{ m/s}^2$$

1. Physical Interpretation

The buoyancy correction reduces the effective surface gravity by a factor of ~ 2.53 . This represents the SCm vacuum medium partially supporting mass against gravitational collapse — the “buoyancy floor” of the UQFF framework.

2. Convergence

The 99-system Γ sweep shows that this solar calibration value is stable across all 7 linewidth values tested ($\Gamma \in \{0.01, \dots, 10.0\}$ THz), confirming that the solar calibration is Γ -independent (as expected for a static surface gravity measurement).

3. Implementation

File: fubi_inside_outside.py, class SolarCalibration147Calc. CP4 class #578.